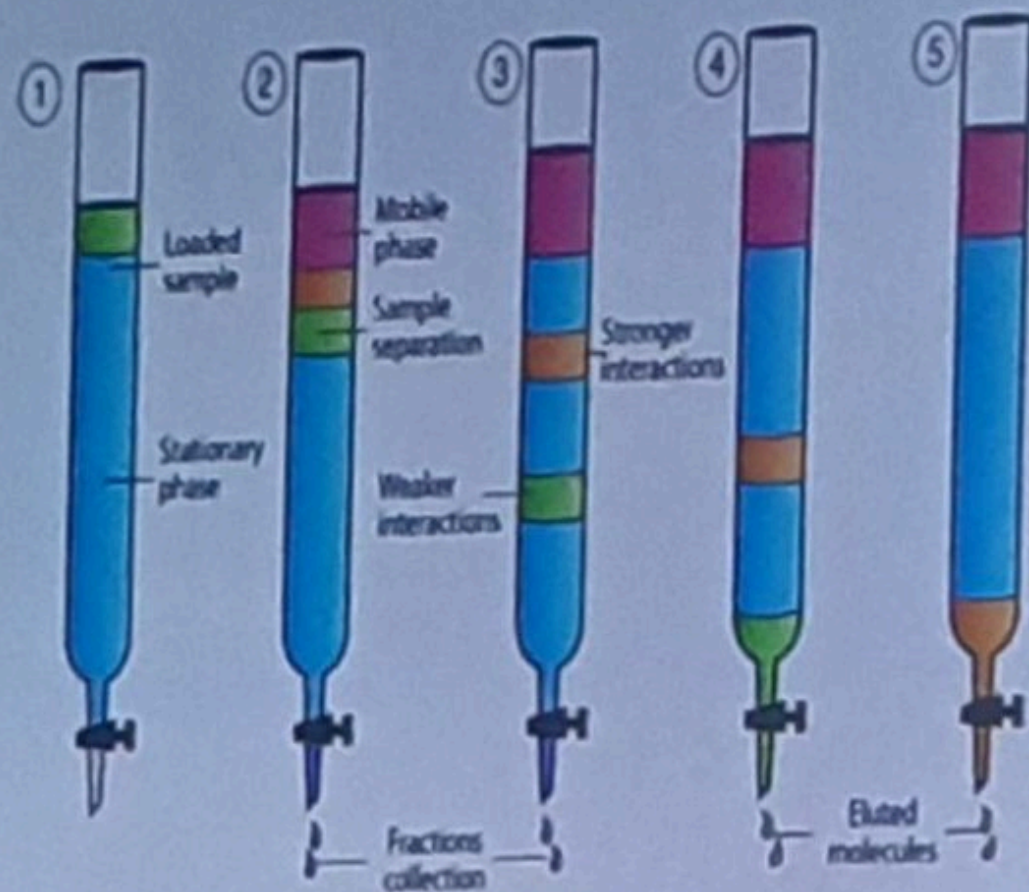
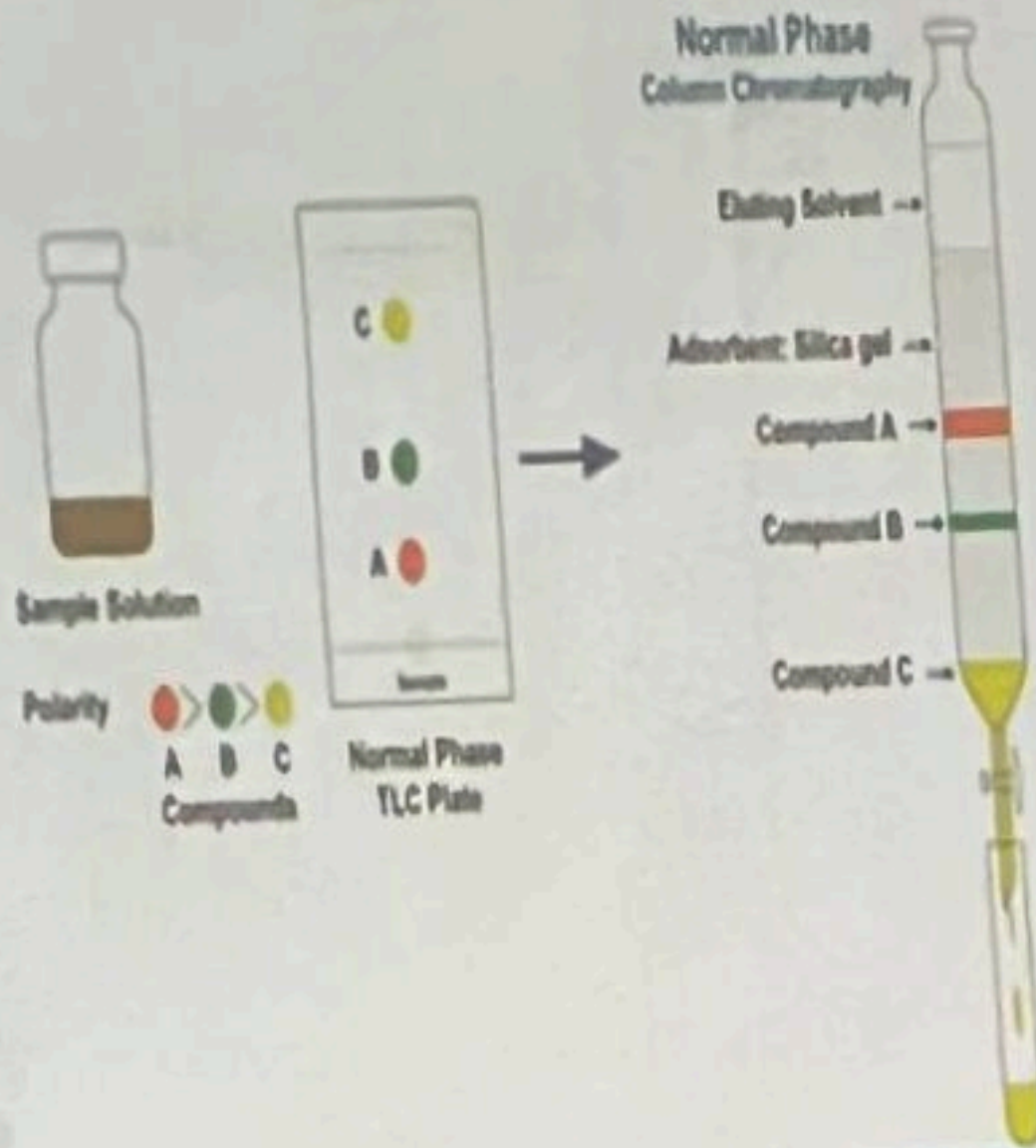


Column Chromatography (CC)

- Column chromatography is a chromatographic set up where the stationary phase is packed in a **slender column**,
- Usually placed in a **vertical position** and the mobile phase is introduced through the upper opening.
- Usually, solvent flow is **assisted by gravity** but sometimes could be assisted by external pressure.
- Has two components



COLUMN CHROMATOGRAPHY



Basic components and setup of CC

A. Stationary phase

- The stationary phases are usually finely ground powders or gels and/or are microporous for an increased surface
 - Eg : Silica gel(normal phase), Reversed phase RP (C10,C18..)
- Ideal properties of adsorbent:
 - Of uniform, finely divided particles (↑ S. area)
 - White / colorless
 - Inert and stable to both the Mp and sample
 - Insoluble in most organic solvents
 - Flowable

B. Mobile phase (eluent)

- The *mobile phase* or *eluent* is either a pure solvent or a mixture of different solvents (not exceeding a mix. Of 3- solvents).
- MP or eluent must first be evaluated prior selection using TLC for effective separation of Compounds
- Usually the flow of MP is through gravity but can also be assisted using a pump (flush chromatography)
 - **NB:** The faster the flow of solvent (MP) the better to decrease side diffusion
Van Deemter's equation.

- Ideal properties of mobile phase
 - **Compatible** with the whole chromatographic separation
 - **Inert**, non – explosive and not highly viscous
 - Neither highly **volatile nor highly viscous**
 - Of relatively **low boiling point**
 - **Non-toxicity**
 - **economic**

C. Packing of the column

- It is the incorporation of the **stationary phase** on to the glass column
- Two types of packing

i. Wet Packing

- The adsorbent is slurred with solvent (**Preferably the first solvent to be used in the separation**) and added to the column in portions.
- The suspending liquid is allowed to follow out slowly but solvent should be above the packing at all times (**to avoid drying**)
- Exclude trapped air To ensure uniform packing
- Most commonly used and Results in a more homogenous packing

ii) Dry Packing

- Solid powder of stationary phase is added in portions, with vibration or tamping between additions so as to give maximum packing density (to avoid air bubbles or cracking)
- The column is there thoroughly washed with the first solvent to wash displaces air, washout impurities and prepares the column for separation and elution.
- Trapped air bubbles leads to poor separation and tailing.

D. Sample loading (application)

- Several ways to load a sample

1. Dissolve the sample in the least volume of the 1st solvent (initial mp).

- The sample is then delivered on the top of column **bed using a long pipette.**

2. Direct deposition (coating) of the sample on to some adsorbent to form a friable powder.

- The powder is then **slurred with a little of the initial MP** at the top of the column bed.

3. Use of paper discs impregnated (loaded) with the sample mixture.

- The dried discs are **placed on the top of just – dry surface,**
- Covered with a very thin layer of the solid support and
- Subsequently wetted with few drops of solvent.

E. Column development modes:

i. Development analysis

- Solutes are separated into well-defined zones by a developer. Development is then stopped before the fastest running component comes out of the column.
- The column is dried and the specific bands are removed and solubilized in a suitable solvent.

ii. Isocratic elution analysis

- In this type liquid chromatogram, an eluant is added continuously, in an isocratic manner to wash down the solutes selectively.
- Isocratic meaning a fixed composition of solvents or polarity.

iii. Gradient elution analysis

- The polarity of the mobile phase is increased slowly:
 - a. Step gradient: Δ in MP composition at predetermined interval.
 - b. Continuous gradient: Δ in MP composition is gradual and continuous.

F. Collection and Visualization of column fractions

- Fraction could be collected **based on fixed volumes or specified times.**
- Similar compounds are bulked together and subjected for further analysis.
- Detecting of the separated bands could be done either in situ (on – column) or out of column

a) On – column detection

- Used in case of colored compounds or those that fluoresce in UV light are evaluated **on the column**

b) Out of column detection

- Column fractions can be **collected**, the **solvent evaporated** and subjected to uv light or color reagents **(TLC)**

G. Factors affecting column efficiency and separation

1. Column dimensions

- As length width ratio $\uparrow \rightarrow$ column efficiency continually improves

Reasons:

- Length and number of theoretical plates $\uparrow$
- Uniform packing is obtained in narrower column

2. Adsorbent

A) Particle size of SP:

- As Particle size ↓ the sorption equilibrium on the column increases and separation improved.
- 10 - 2000 μm

B) Uniformity of packing of the column

- Efficient separation if uniform Packing

C) Adsorbent activity

- water deactivation of silanol group

3. Mobile phase / solvent/

- **selection of the solvent**

- flow rate is inversely proportional to viscosity (**low viscosity is better**)
- Solvent/solvent system (**not exceeding a mix. Of 3– solvents**)
- Solvents can be grouped in an increasing or decreasing order of elution strength (**polarity**)

- **Constancy of flow**

- The flow should be **continuous**.

- **Solvent flow rate**

- **Uniform and moderate flow rate** will result in a more satisfactory separation

4. Concentration of Solute

- At high conc., the **solute band over shoots and moves more rapidly** through the column "Bearding" or "Tailing"
- Sample to adsorbent ratio of 1:30 to 1:50 should be used.

5. Temperature

- The chromatographic operation should be done at **constant temperature**, because temperature affects:
 - **Distribution co-efficient** of the components
 - **Viscosity of the solvent** -> flow rate
 - **Adsorption** is also ↓ at higher temperature.

Advantage

- Complex mixtures of compounds can be separated
- Large amount of sample can be separated
- Flexible MP composition

Disadvantage

- Time consuming
- High solvent consumption